Light

GCSE Computer Science – Network Protocols (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Without network protocols, what problem would occur when two computers try to communicate?

- A. The computers would not understand each other's data formats
- B. The computers would communicate too quickly
- C. The computers would use too much electricity
- D. The computers would automatically disconnect from the internet

Q2 (1 mark)

A student researches homework topics using a web browser. Which protocol enables the browser to retrieve web pages from websites?

- ○ **A.** SMTP enables the browser to retrieve web pages
- ○ **B.** IMAP enables the browser to retrieve web pages
- ○ **C.** HTTP or HTTPS enables the browser to retrieve web pages
- ○ **D.** TCP alone enables the browser to retrieve web pages

Q3 (1 mark)

Why do online shops use HTTPS instead of HTTP when customers enter payment details?

- ○ **A.** HTTPS makes the website load faster for customers
- ○ **B.** HTTPS is free while HTTP requires payment
- ○ **C.** HTTPS only works with credit cards, not debit cards
- ○ **D.** HTTPS encrypts payment details so they cannot be read if intercepted

Q4 (1 mark)

How does TCP ensure that data packets arriving out of order can still be correctly reassembled?

- ○ **A.** TCP assigns sequence numbers to packets so they can be reordered
- ○ **B.** TCP encrypts each packet with a unique password
- ○ **C.** TCP sends all packets through the same physical cable
- ○ **D.** TCP waits until all packets arrive before sending any

Q5 (1 mark)

What does an IP address identify on a network?

- ○ **A.** The speed of the internet connection
- ○ **B.** The brand of computer being used
- ○ **C.** The password required to access the network
- ○ **D.** A specific device or location on the network

Q6 (1 mark)

A photographer emails a client to confirm a booking. Which protocol handles the transmission of this outgoing email?

- ○ **A.** HTTP handles the outgoing email transmission
- ○ **B.** IMAP handles the outgoing email transmission
- ○ **C.** HTTPS handles the outgoing email transmission
- ○ **D.** SMTP handles the outgoing email transmission

Q7 (1 mark)

An estate agent reads an email on their phone, then later opens the same inbox on their office computer. The email shows as 'read' on both devices. Which protocol makes this synchronisation possible?

- ○ **A.** IMAP, because it keeps emails on the server and synchronises status across devices
- ○ **B.** SMTP, because it sends the read status to all devices

- ○ **C.** HTTP, because it transfers the email between devices
- ○ **D.** TCP, because it creates reliable connections

Q8 (1 mark)

Web browsers and email programs operate at which layer of the TCP/IP model?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q9 (1 mark)

Which TCP/IP layer is responsible for setting up communication between hosts and ensuring reliable data delivery?

- ○ **A.** Transport layer
- ○ **B.** Application layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q10 (1 mark)

Which TCP/IP layer handles addressing data packets and routing them across networks?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q11 (1 mark)

Network interface cards (NICs) and device drivers operate at which TCP/IP layer?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q12 (1 mark)

What does IP determine when sending data across the internet?

- ○ **A.** The colour scheme of web pages
- ○ **B.** The password required for email accounts
- ○ **C.** The speed at which users can type
- ○ **D.** The route that data packets take across networks

Q13 (2 marks)

A veterinary clinic's system transmits patient records to a specialist. How does TCP confirm that data has been successfully received?

- ○ **A.** TCP assumes all data arrives correctly and does not check
- ○ **B.** TCP sends the data three times and hopes at least one copy arrives
- ○ **C.** The receiver sends acknowledgment messages back to confirm receipt, and TCP retransmits any unacknowledged packets
- ○ **D.** TCP converts all data to images which are easier to transmit

Q14 (2 marks)

A music streaming service delivers songs from servers in Sweden to listeners in Australia. What addressing information does IP add to each data packet?

- ○ **A.** The name of the song and the artist
- ○ **B.** The source IP address (Sweden server) and destination IP address (Australian listener's device)
- ○ **C.** The username and password of the listener
- ○ **D.** The file size and audio quality settings

Q15 (2 marks)

A customer browses products on an online furniture shop. Describe how HTTP operates when the customer clicks on a product to view its details.

- ○ **A.** HTTP sends the product directly to the customer's home

- ○ **B.** HTTP encrypts the customer's personal information
- ○ **C.** The browser sends an HTTP request to the server, which responds with the product page content
- ○ **D.** HTTP converts the product image into text for faster loading

Q16 (2 marks)

A hospital website has a public information section and a patient portal for viewing medical records. Which protocol should be used for each section?

- ○ **A.** HTTP for both sections since they are on the same website
- ○ **B.** HTTPS for the patient portal only; HTTP is acceptable for public information
- ○ **C.** HTTPS for both sections to ensure all communications are encrypted
- ○ **D.** Neither protocol is needed since hospitals have special medical protocols

Q17 (2 marks)

A teacher sends homework instructions to students via email, and students reply with completed work. Explain the roles of SMTP and IMAP for the teacher.

- ○ **A.** SMTP handles both sending and receiving for the teacher
- ○ **B.** IMAP sends the homework instructions; SMTP receives the student replies
- ○ **C.** SMTP sends the teacher's homework instructions; IMAP retrieves the student replies from the server

- ○ **D.** IMAP handles all email functions; SMTP is obsolete

Q18 (2 marks)

HTTP, HTTPS, SMTP, and IMAP all operate at the application layer. What characteristic do they share that places them at this layer?

- ○ **A.** They all handle physical signal transmission through cables
- ○ **B.** They all assign IP addresses to devices
- ○ **C.** They all provide services directly to user-facing applications like browsers and email programs
- ○ **D.** They all break data into packets for transmission

Q19 (2 marks)

Before data transfer begins, the transport layer negotiates settings between the sending and receiving hosts. What is one setting that is agreed upon?

- ○ **A.** The colour scheme for displaying the data
- ○ **B.** The physical type of cable to use for transmission
- ○ **C.** The size of data packets to be used during transmission
- ○ **D.** The brand of computer at each end

Q20 (2 marks)

A video call connects users in Canada and New Zealand. Data packets travel through multiple intermediate networks. What does the internet layer do at each intermediate point?

- ○ **A.** It examines destination addresses and forwards packets toward New Zealand
- ○ **B.** It displays the video to administrators at each point
- ○ **C.** It stores copies of the video for later viewing
- ○ **D.** It converts the video into audio format

Q21 (2 marks)

When data reaches the link layer for transmission across a wired network, what conversion takes place?

- ○ **A.** Text is converted into images for easier reading
- ○ **B.** Digital data is converted into electrical signals for transmission through cables
- ○ **C.** Email addresses are converted into phone numbers
- ○ **D.** Web pages are converted into downloadable files

Q22 (2 marks)

An architect emails large building plans to a construction company overseas. Explain the distinct roles of TCP and IP in this transmission.

- ○ **A.** TCP and IP both handle addressing—they perform identical functions
- ○ **B.** IP breaks the plans into packets; TCP determines the route
- ○ **C.** TCP ensures reliable delivery of all data; IP handles addressing and routing across international networks

○ **D.** IP encrypts the building plans; TCP compresses them for faster sending

Q23 (2 marks)

When a user accesses a website, protocols at multiple TCP/IP layers work together. Which statement correctly describes this cooperation?

○ **A.** Only one protocol is needed—the others are redundant

○ **B.** All protocols perform the same function but at different speeds

○ **C.** Protocols at higher layers can replace those at lower layers

○ **D.** Each layer's protocol handles a specific task: HTTP requests the page, TCP ensures reliable delivery, IP routes packets, and the link layer transmits signals

Q24 (3 marks)

A journalist submits an article to their editor via a web form. Describe how data flows through all four TCP/IP layers when the article is submitted.

○ **A.** Link → Internet → Transport → Application: the article starts at the cable and moves up

○ **B.** Application → Transport → Internet → Link: the web form uses HTTPS to package the article, TCP segments it for reliable delivery, IP adds addresses and determines routing, the NIC transmits signals

○ **C.** Transport → Application → Link → Internet: TCP must process data before any application can use it

○ **D.** The article bypasses the layers and goes directly to the editor

Q25 (3 marks)

A solicitor's office sends important legal documents to the court electronically. Explain all the mechanisms TCP uses to ensure these documents arrive complete and without errors.

- ○ **A.** TCP only checks that files are the correct format; it does not verify content
- ○ **B.** TCP assigns sequence numbers for correct ordering, sends acknowledgments to confirm receipt, performs error checking on packets, and automatically retransmits any lost or corrupted data
- ○ **C.** TCP sends documents three times and uses whichever copy arrives first
- ○ **D.** TCP relies entirely on the email system to verify document delivery

Q26 (3 marks)

A recruitment agency sends interview invitations to candidates and receives confirmation replies. The agency uses multiple devices to manage this correspondence. Describe the complete role of email protocols in this workflow.

- ○ **A.** A single protocol handles all email functions regardless of direction
- ○ **B.** SMTP transmits all outgoing interview invitations to email servers; IMAP retrieves confirmation replies from the server and synchronises them across all agency devices because emails remain server-stored
- ○ **C.** Emails are transmitted directly between devices without using servers
- ○ **D.** IMAP sends invitations because it is newer; SMTP receives replies because it is faster

Q27 (3 marks)

A global charity organisation coordinates disaster relief between offices in Kenya, Philippines, and Brazil. Explain how IP enables communication between these geographically dispersed locations.

- ○ **A.** IP requires dedicated cables connecting each pair of countries directly
- ○ **B.** IP assigns unique addresses to each office's systems, adds source and destination addresses to packets, and routes data through multiple intermediate networks regardless of physical distance
- ○ **C.** IP only works within a single continent; different protocols connect continents
- ○ **D.** IP broadcasts all data to every country simultaneously

Q28 (3 marks)

An online therapy service handles extremely sensitive mental health conversations. Evaluate why HTTPS is essential and what protection it provides throughout the user's session.

- ○ **A.** HTTPS is optional for therapy services since conversations are not financial
- ○ **B.** HTTPS encrypts all session data during transmission, protecting login credentials, personal details, and sensitive conversations from interception, ensuring confidentiality throughout the therapeutic relationship
- ○ **C.** HTTPS only protects the initial login; conversations after that are unencrypted
- ○ **D.** HTTPS replaces the need for any other security measures

Q29 (3 marks)

A graphic designer downloads high-resolution images from a stock photo website. Describe how the image data passes through the TCP/IP layers on the designer's computer when being received.

- ○ **A.** Application → Transport → Internet → Link: data flows downward when receiving
- ○ **B.** Link → Internet → Transport → Application: the NIC receives signals, IP verifies the destination address, TCP reassembles packets in order, the browser application displays the images
- ○ **C.** All layers receive data simultaneously to maximise speed
- ○ **D.** Only the application layer is involved in receiving downloads

Q30 (3 marks)

A cinema's online booking system allows customers to select seats, enter payment details, and receive email confirmations. Identify which protocols handle each part of this process.

- ○ **A.** HTTP handles everything since it is all one website
- ○ **B.** HTTPS handles the web interface and payment processing with encryption; SMTP sends the email confirmation from the cinema's server to the customer
- ○ **C.** IMAP handles seat selection; TCP handles payments; SMTP handles confirmations
- ○ **D.** Each step requires a completely different set of protocols with no overlap

Total: **55** marks • Time: **30 min**

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